Artificial Intelligence

Concept Notes in Simple Language

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Covers: what intelligence means, the four views of AI, the foundations and history of AI, intelligent agents and PEAS, rationality, the properties of task environments, the five agent types, and a guide to applying these ideas to scenario-based questions.

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Part 1. What AI Is: Intelligence and the Four Views

1.1 What do we mean by intelligence?

Before we can build artificial intelligence, we need to ask what intelligence is. The slides make one important point: human intelligence is not a single ability. It is a bundle of several different capabilities that work together. If you imagine a person solving a real problem, such as finding their way to a new place, they use many of these abilities in a few seconds.

- **Perception** is recognizing objects, sounds, faces or situations. A person who sees a red light and understands it is a traffic signal is using perception.
- **Learning** is improving performance with experience. The more you practise, the better you become.
- **Reasoning** is drawing conclusions from the information you have. If all birds have wings and a sparrow is a bird, then a sparrow has wings.
- **Problem-solving** is finding solutions to new situations that you have not seen before.
- **Planning** is deciding which actions to take and in what order, for example the steps needed to cook a meal.
- **Language ability** is understanding and producing communication.
- **Adaptation** is changing your behaviour when circumstances change, such as taking another road when the usual one is blocked.
- **Decision-making** is choosing between alternatives.

AI systems try to reproduce *some* of these abilities using computation. This is the key difference between humans and machines. A human mind is flexible, general-purpose and emotionally aware, and it learns from life experience. Most AI systems are strong at specific tasks, learn from data, have no real emotions, and are often narrow in scope. A chess program may beat every human, yet it cannot make a cup of tea. So an AI system can be extremely capable in one area and very weak in another.

> ***Key idea:*** *Intelligence is a collection of abilities. AI copies some of them, not all of them.*

1.2 Four classical views of artificial intelligence

Over the years, researchers have defined AI in four different ways. They come from asking two simple questions. First, is the goal to imitate **humans** or to be **rational**, meaning to do the logically best thing? Second, are we looking at how a machine **thinks** or at how it **acts**? Combining the answers gives four views.

Thinking humanly. The question is whether a machine can think in the same way that humans think. Work in this view tries to model human mental processes such as memory, learning and reasoning. It overlaps strongly with psychology and neuroscience.

Acting humanly. The question is whether a machine can behave like a human. The famous test here is the Turing Test: if a person talks to a machine and cannot tell whether it is a machine or a human, the machine passes. Chatbots and humanoid robots belong to this view. Notice that this view only cares about outward behaviour, not about how the machine thinks inside.

Thinking rationally. The question is whether a machine can reason correctly using the laws of logic. For example, from the facts "If A then B" and "A is true" the machine concludes "B is true". This view uses logic, mathematical proof and knowledge representation.

Acting rationally. The question is whether a machine can choose the actions that best achieve its goals. Such a system sets a goal, evaluates its options, chooses the best action and achieves the goal. Autonomous cars, game-playing AI and intelligent agents are examples.

The first two views are called **human-centred** approaches and the last two are **rational** approaches. Modern AI mainly focuses on **rational agents**, although ideas from all four views remain useful. This is why the rest of the course is about agents: a rational agent is simply something that does the right thing given what it knows.

> ***Key idea:*** *Human-centred approaches ask, "Is it like us?" Rational approaches ask, "Does it do the best thing?" Modern AI mostly asks the second question.*

Part 2. Where AI Comes From: Foundations and History

2.1 Intellectual foundations: AI as a meeting point of many fields

AI did not grow out of computer science alone. It borrowed important ideas from many subjects, which is why it is called an interdisciplinary field. Each subject contributes a different piece of the puzzle.

Philosophy asks the deepest questions: What is knowledge? How do humans reason? What is consciousness? Can machines possess intelligence? What is the relationship between the mind and the physical brain? Philosophy gave AI the idea that reasoning can be treated as a systematic process following formal rules, and it keeps raising questions about ethics, such as who is responsible when an intelligent system makes a harmful decision. The mind-body problem and the question of whether a machine can be truly conscious or only simulate consciousness are still open.

Mathematics supplies the tools for representing and manipulating information. Each branch has a clear job. *Logic* is used for rule-based reasoning, such as "if temperature is above 38 and cough is present, then possible infection". *Probability* helps a system reason under uncertainty. Instead of a plain yes or no, a medical system can say the probability of a disease given the symptoms is 0.82. *Statistics* finds patterns in data, for example learning the relationship between house size and price in order to predict the price of a new house. *Linear algebra* is essential for machine learning because data such as images and words are stored as vectors and matrices. An image, for instance, is a grid of numbers where each number is the brightness of one pixel. *Optimization* finds the best parameter values, meaning the ones that give the least error. A common method is gradient descent, which you can picture as walking downhill on a curve until you reach the lowest point, where the error is minimum.

Economics contributes ideas about rational decision-making: utility (how valuable an outcome is), decision theory, game theory, resource allocation and incentives. The core idea is to compare cost and benefit and choose the option with the highest expected value.

Neuroscience studies the brain and nervous system and inspired neural networks. It is explained in more detail in the next subsection.

Psychology gives insight into human perception, memory, problem solving, learning and behaviour.

Computer engineering provides the hardware, such as fast processors, GPUs, sensors and embedded systems, without which AI could not run.

Control theory gives methods to design systems that achieve a desired behaviour automatically, using feedback. A drone that keeps itself stable follows the cycle of sense, control, act. This feedback idea is closely related to how agents work.

Linguistics provides knowledge about the structure and meaning of language: syntax, semantics, pragmatics, language understanding and language generation.

2.2 Neuroscience and the artificial neuron

The human brain contains billions of neurons connected in complex networks. A biological neuron has dendrites that receive signals, a cell body that processes them, an axon that carries the signal, and axon terminals that send it on to other neurons.

An artificial neuron is a very simplified mathematical copy of this. It receives several inputs x_1, x_2, up to x_n. Each input has a weight w_1, w_2, up to w_n, which says how important that input is. The neuron computes a weighted sum plus a bias:

$$z = (w_1x_1 + w_2x_2 + ... + w_nx_n) + b$$

The bias b is a constant that shifts the result up or down. The value z is then passed through an **activation function** f, which produces the final output y = f(z). The three steps are therefore: compute the weighted sum, apply the activation function, produce the output. Think of it like deciding whether to go out: you weigh several factors (weather, mood, homework), each with its own importance, add them up, and then decide yes or no.

An important distinction: artificial neural networks are *inspired by* the brain, but they are highly simplified models. They capture the idea of learning from examples, but they do not replicate consciousness, emotions or the detailed biology of the brain.

2.3 Computer engineering and why GPUs matter

Training a neural network involves millions or billions of matrix multiplications, written as C = A x B. A CPU works mostly in sequence, doing a few operations at a time, which suits general tasks. A GPU has thousands of small units that perform many calculations at the same moment, called parallel processing. Because matrix operations can be split into many independent calculations, GPUs make deep learning much faster. Other advances also helped: specialized accelerators such as TPUs, larger memory, cloud computing, distributed computing, high-speed networks and large-scale storage. Better hardware leads to more capable AI, which leads to real-world applications at scale.

2.4 Linguistics and natural language processing

Linguistics is the scientific study of language: grammar (rules for forming correct sentences), syntax (how words are arranged), semantics (the meaning of words and sentences), context (how surrounding information changes meaning), and speech and communication. These ideas created the field of **Natural Language Processing (NLP)**, which enables computers to understand, interpret and generate human language. Its applications include machine translation, speech recognition, chatbots, question answering, text summarization, search engines and generative AI assistants. Modern language models learn language patterns directly from extremely large collections of text, and can understand language, generate text, follow instructions and hold conversations.

2.5 The Dartmouth Conference and the birth of AI

The term **Artificial Intelligence** is strongly linked to the Dartmouth Summer Research Project held in 1956 at Dartmouth College in the USA. John McCarthy coined the term. Other key figures were Marvin Minsky (AI and cognitive science), Claude Shannon (information theory), Allen Newell (problem solving and cognitive modelling) and Herbert Simon (decision-making and symbolic reasoning). The proposal was to carry out a two-month study of AI by ten people in the summer of

1956. Researchers were very optimistic and believed that human intelligence could be described precisely enough for a machine to simulate it.

A short timeline helps: before the 1950s there were ideas about mechanical reasoning; in 1950 Alan Turing published "Computing Machinery and Intelligence"; in 1956 the Dartmouth Conference named the field; and in the late 1950s the first AI programs appeared. Dartmouth is considered the formal beginning of AI as an academic field because it introduced the name, gathered key researchers and set the research agenda.

> ***Key idea:*** *AI is a meeting point of many fields, and its formal birth as a field was the 1956 Dartmouth Conference.*

2.6 Expert systems

In the late 1970s and 1980s, expert systems tried to capture the knowledge of human specialists as rules. An expert system has two main parts. The **knowledge base** stores facts and rules about one domain. The **inference engine** applies those rules to the information it receives and derives conclusions. A simple medical rule might say: IF fever is high AND cough is present AND fatigue is present, THEN investigate a respiratory infection.

It works in three steps: the system receives information such as symptoms, the inference engine checks which rules match, and if a rule matches the system gives a conclusion or recommendation. They were used in medicine, engineering, finance, manufacturing and equipment diagnosis. They proved AI could be commercially valuable and brought funding and renewed optimism. However, they worked only in narrow domains, the knowledge had to be written in by hand, they handled uncertainty and common sense poorly, and they performed badly outside their specific area.

2.7 Why machine learning became successful

Machine learning succeeded because four things came together. First, there was **more digital data** from the internet, smartphones, online services, sensors and social networks. Second, there was **greater computing power**, as processors became faster and cheaper and the cloud made high performance accessible. Third, there were **better algorithms**: classification (for example spam or not spam), regression (predicting numbers such as house prices), clustering (grouping similar data such as customer segments), probabilistic modelling (handling uncertainty) and neural networks. Fourth, there was **improved storage**, which made it practical to keep and process huge datasets. Notice the difference from expert systems: instead of humans writing the rules, the machine learns the patterns from data.

2.8 The deep-learning era

Deep learning is a branch of machine learning that uses neural networks with many layers; the word deep refers to this large number of layers. Each layer transforms its input into a slightly more useful representation. In image recognition, early layers detect simple things such as edges, colours and textures. Middle layers combine them into shapes, patterns and object parts. Later layers recognize complete objects such as faces, cars and animals. The network builds understanding step by step, from simple to complex, and it learns these features automatically without engineers designing them by hand.

Deep learning became successful because of large datasets, powerful computing, improved algorithms and better tools such as TensorFlow and PyTorch. Its impact includes better image and speech recognition, breakthroughs in language processing such as large language models, and wide use in healthcare, finance and robotics.

2.9 Generative AI

Traditional predictive AI answers a specific question about existing data. Examples are "Is this transaction fraudulent?", "What will tomorrow's demand be?" and "Which product should be recommended?". It predicts or classifies, but does not create anything new. **Generative AI** creates new content such as text, images, audio, music, video, computer code and designs. If you ask it to write a professional email requesting a meeting next Monday, it produces a complete email. This matters because people can interact with AI in natural language, use it creatively, and apply it in education, business, healthcare and entertainment.

2.10 AI agents and tool use

Modern AI research increasingly focuses on **AI agents**. An agent understands a goal, plans a sequence of steps, takes actions (often using external tools), and observes the results so it can adjust. It does more than produce one prediction; it works toward a goal through many steps. The typical cycle is: goal, plan, action, observation, adjustment, repeated until the goal is achieved. Tools include web search, databases, calculators, code editors, APIs, computer applications and even robots.

For example, given the goal "Plan a three-day business trip", an agent determines the requirements, searches transport options, compares hotels, organizes a schedule and produces an itinerary. The benefits are that it can finish complex multi-step tasks and increases productivity. The challenges are that it may still make mistakes, needs human oversight, carries security and privacy risks, and needs clear goals and constraints.

Part 3. Intelligent Agents and PEAS

3.1 What is an agent?

An **agent** is anything that **perceives** its environment through **sensors** and **acts** upon that environment through **actuators**. The flow is: Environment, then Sensors, then Agent, then Actuators, then back to the Environment. A human is an agent (eyes and ears are sensors, hands and legs are actuators). A robot, a self-driving car, a software assistant and a vacuum-cleaning robot are also agents. The textbook for this course, AIMA (Artificial Intelligence: A Modern Approach), introduces agents in exactly these terms.

3.2 Percepts and percept sequences

A **percept** is the information an agent receives at one particular moment. For a vacuum agent, a percept might be: Location = Room A, Dirt detected = Yes. A **percept sequence** is the complete history of everything the agent has perceived so far. For example, at time t1 the vacuum is at A and finds dirt; at t2 it is at A and the dirt is gone; at t3 it is at B and finds dirt; at t4 it is at B and the dirt is gone.

An agent can choose its next action using the current percept, previous percepts, built-in knowledge and learned information. AIMA describes an agent's behaviour as a mapping from percept sequences to actions, which is called the **agent function**. In plain words: given everything the agent has seen so far, what will it do next?

3.3 AI model versus AI agent

These two terms should not be confused. An **AI model** mainly performs a computation: it takes an input and produces an output. If a student asks "Explain binary search", the model generates an explanation and stops. An **AI agent** can perceive, reason, act and use tools to reach goals in an environment. An AI coding agent, for instance, reads code files and terminal output, edits files, runs commands and runs tests.

3.4 The modern agent loop

Today's agents follow a repeating loop. Take the goal "Fix the login bug." The agent **perceives** by reading source files, error messages and test results. It **decides** (for example, "I should inspect the authentication function"). It **acts** by opening the file, changing code and running the test. It **observes** the result ("Test failed"), and then **decides again** using this new information. The process repeats until the goal is achieved. The short form is Observe, Reason, Act, then Observe, Reason, Act again. An agent is therefore more than simple text generation: it observes, reasons, uses tools, observes the results and continues.

3.5 Actuators in modern agents

An **actuator** is whatever allows an agent to change its environment. A physical agent such as a robot uses wheels, motors and robotic arms. A software agent's actuators include writing a file, editing code, running a terminal command, searching the web, calling an API, sending a message, filling a web

form and clicking a button. Computer-using agents can operate graphical interfaces, opening applications, clicking buttons and filling forms just like a human user. In every case, actuators are how the agent changes its environment.

3.6 PEAS: describing an agent's task

PEAS is a checklist used to describe the task an agent must solve. It stands for four things.

- **Performance measure** answers what "good" means. It is how success is judged.
- **Environment** answers where the agent works.
- **Actuators** answer what the agent is allowed to do.
- **Sensors** answer what information the agent can access.

The slide example is an AI research agent given the task "Compare solar and wind power using recent reliable information." Its performance measure is correct information, relevant sources, recent information, clear explanation and proper citations. Its environment is the internet, search engines, websites, research papers and documents. Its actuators are searching, opening pages, reading documents, analysing information and creating a report. Its sensors are search results, webpage text, tables, documents and user instructions. The work flows as: search, read, analyse, create report.

> ***Key idea:*** *A common mistake is mixing up performance measure and actuator. The performance measure is how we judge success (for example, delivering correctly and safely). The actuator is a physical or software means of acting (for example, wheels or a sorting gate). Sensors are for input; actuators are for output.*

Part 4. Rationality, Tools and Verification

4.1 Rationality versus omniscience

A **rational agent** makes the best decision it can using the information it has at that moment. An **omniscient agent** would somehow know everything that will happen in the future. Real agents are never omniscient; they work under uncertainty.

The slide example shows a self-driving agent. Traffic data says Route A takes 12 minutes, Route B takes 18 and Route C takes 25, so the rational choice is Route A. If an accident then happens on Route A seven minutes later, the agent was not irrational; it simply could not have known. An omniscient agent would have known about the accident beforehand. Therefore rationality is not the same as knowing the future. We judge an agent by whether its choice was sensible given its knowledge, not by whether luck turned out well.

4.2 Chatbot versus research agent

Suppose we ask, "What is Pakistan's current inflation rate?" A simple chatbot answers from information it learned earlier, so it may give a confident but outdated number. An agent understands that current data is needed, searches the web, finds a reliable source, reads the data, compares sources if necessary and gives an answer with evidence. The lesson is that **sometimes using a tool is itself the rational action**. A good agent decides when it should look something up instead of guessing.

4.3 AI can write code, but can you verify it?

AI can generate hundreds of lines of code in seconds, but someone must still answer important questions. Is the code correct? Is it secure? Is it maintainable? Does it meet the requirement? Does it break existing features? What happens at large scale? Can we trust its dependencies? Are there hidden edge cases? Verification, evaluation and judgment remain human responsibilities. A valuable modern skill is therefore not only generating code but reading and understanding AI-generated code, checking correctness, security and reliability, testing edge cases, and improving or refactoring when needed. Verification is not optional; it is an essential engineering skill.

Part 5. Properties of Task Environments

Two questions guide agent design. The first is, *what kind of world does the agent operate in?* (the nature of the environment). The second is, *what information should the agent use to choose actions?* (the structure of the agent). The environment shapes the design: the environment often determines the kind of intelligence the agent needs. To describe an environment we classify it along seven dimensions, each as a pair or trio of choices. Below, each one is explained with a simple test question you can ask yourself.

5.1 Fully observable or partially observable

Test question: can the agent see everything relevant to its decision? In a **fully observable** environment the sensors give all the information needed to choose an action. A chess board, a crossword puzzle and a simple vacuum world with complete sensors are examples. In a **partially observable** environment, important information is hidden, missing, uncertain or noisy. Examples are a self-driving vehicle (it cannot see around corners), a robot using cameras, poker (you cannot see the other players' cards) and a research agent browsing incomplete web information. A partially observable agent may need memory or an internal model to keep track of what it cannot see right now.

5.2 Single-agent or multiagent

Test question: do other decision-makers affect the outcome? In a **single-agent** environment only one relevant decision-maker exists, such as solving a crossword, classifying an image or solving a mathematical puzzle. In a **multiagent** environment other agents also choose actions that affect the outcome. Examples are chess, traffic, online auctions, robot teams and several AI agents collaborating on software. Multiagent environments may be **cooperative**, **competitive** or **mixed**. Chess is competitive, while traffic has both cooperative and competitive aspects. In the modern multi-agent AI example, one agent builds the backend, another the frontend, another writes tests, another reviews security and another coordinates. They must communicate, divide tasks, share context, detect conflicts and coordinate changes, so the environment becomes more complex.

5.3 Deterministic, stochastic or strategic

Test question: does the same action in the same situation always give the same result? In a **deterministic** environment, the current state and action completely determine the next state. In the simple vacuum world, if the agent chooses "suck" the dirt is always removed. In a **stochastic** environment the same action may produce different outcomes, so the result is a probability distribution over possible next states. Driving, robot grasping (which may fail due to slipping), medical treatment (different effects on different patients) and financial markets are stochastic. In a **strategic** environment the uncertainty mainly comes from another agent's choices, as in chess or traffic. AIMA uses the word strategic when uncertainty is mainly due to other agents.

5.4 Episodic or sequential

Test question: does today's decision affect tomorrow's situation? In an **episodic** environment each decision is mostly independent of previous ones: percept, action, and the episode ends. Classifying one image, detecting whether one email is spam, and inspecting individual manufactured parts are

episodic. In a **sequential** environment the current action influences future situations, so the agent must consider long-term consequences and not just the immediate next action. Chess, driving, planning, software modification and robot navigation are sequential. Sequential problems need planning and search and are generally harder. As the slides say, classification AI is often episodic, while agentic AI for real-world tasks is sequential.

5.5 Static, dynamic or semidynamic

Test question: does the world change while the agent is still thinking? In a **static** environment nothing changes while the agent decides, so it can take as much time as it needs; a crossword puzzle is an example. In a **dynamic** environment the world may change while the agent is reasoning, so decisions must be made in real time. Driving, stock trading, robot navigation and cybersecurity monitoring are dynamic. In a **semidynamic** environment the environment itself does not change, but the performance score changes with time. Chess with a clock is the classic example: the board stays the same, but taking too long reduces your performance.

5.6 Discrete or continuous

Test question: are the states and actions separate countable steps, or smooth values? A **discrete** environment has a finite or clearly separated set of states and actions: chess moves, tic-tac-toe and menu selections. A **continuous** environment has states and actions that can take any value in a range, such as vehicle speed, a robot's joint angle, distance and temperature. Real physical environments such as driving and robotics are largely continuous, even if we sometimes approximate them as discrete. Continuous environments require sensing, estimation, precise control and often real-time handling.

5.7 Known or unknown

Test question: does the agent know the rules of how its actions change the world? In a **known** environment the agent knows how its actions affect the environment, so it can predict outcomes. Chess, the simple vacuum world and a mathematical problem are known. In an **unknown** environment the agent must discover how the world behaves by interacting with it, such as a robot in a new building. This is different from observability. **Observable** asks, "What is happening now?" **Known** asks, "Do I understand how this world works?" A robot can clearly see a corridor (observable) yet not know where it leads (unknown). Unknown environments require exploration and learning.

5.8 Case studies from the slides

AI coding agent (task: find and fix an authentication bug). It is partially observable because requirements and system details may be missing. It is sequential because each code change affects later tests. It is partly stochastic because dependencies and external services may vary. Its actions are mostly discrete (edit, test, search, commit). It is dynamic because the repository or services may change while it works. It is potentially multiagent because humans or other agents can modify the code. The point is that real-world agents combine several of these properties at once.

AI research agent (task: produce an evidence-based report). It is partially observable because not all information is accessible, sequential because each step affects later ones, dynamic because web content keeps changing, and uncertain or stochastic because sources may be incomplete, conflicting or

unreliable. The agent must decide which sources to trust.

Part 6. The Five Agent Types

AIMA presents a progression of agent designs, from very simple to more capable. More capability usually comes with more complexity. The first four are basic agent-program families; learning can be added to any of them.

6.1 Simple reflex agent

This agent responds directly to the **current percept** using condition-action rules, which are just if-then rules, and it ignores history. A thermostat (if the temperature is below the set value, switch on the heater) and a very simple vacuum cleaner (if dirty, suck) are examples. It works well only when the current percept contains everything needed, meaning the environment is fully observable and the decision depends only on the present situation.

6.2 Model-based reflex agent

This agent still uses condition-action rules, but it also keeps an **internal state**, which is its belief about the world. This lets it handle **partial observability**. A robot vacuum that keeps a map, an autonomous drone that tracks its state, and a game agent that remembers unseen parts of the world are examples. The key idea is to use an internal model to represent the world, not just the current percept. It still reacts using rules; it simply knows more about the situation.

6.3 Goal-based agent

This agent has a **goal** and considers the **future consequences** of its actions. It uses search or planning to find a sequence of actions that leads to the goal. GPS route planning, robot navigation to a target and an AI coding agent fixing a bug are examples. The key idea is to choose actions that help achieve a goal, possibly by planning ahead. A goal only says whether a state is acceptable or not; it does not say which acceptable state is better.

6.4 Utility-based agent

When there are many ways to reach a goal, or trade-offs between competing aims, a goal alone is not enough. A **utility-based agent** uses a **utility function**, a numerical score of how good each outcome is. It evaluates the possible actions and picks the one with the **highest expected utility**, balancing several objectives. Examples are a self-driving car balancing safety, time and comfort, a recommendation system balancing relevance and diversity, and resource management balancing cost and performance. Because outcomes are uncertain, the agent multiplies each outcome's value by its probability and adds them up; this is the expected utility.

The difference from a goal-based agent is simple: a goal-based agent asks, "Does this reach the goal?" (yes or no), while a utility-based agent asks, "How good is this outcome, and how likely is it?" Maximizing expected utility does not guarantee a good result every time; it gives the best choice on average given the uncertainty.

6.5 Learning agent

A learning agent **improves its behaviour through experience**. It has four parts. The **performance element** chooses the actions. The **critic** evaluates how well the agent is doing, using a performance standard such as rewards or feedback. The **learning element** uses that feedback to improve the performance element. The **problem generator** suggests new, exploratory actions so the agent can gain useful experience. Examples include LLM-based agents that learn from user feedback, game-playing agents such as AlphaGo, and robots that learn through trial and error. Learning can be combined with any of the other four types. In practice, modern systems often combine goal-based, utility-based and learning ideas.

> ***Key idea:*** *Remember the progression with one question each. Simple reflex: what do I see now? Model-based: what do I see now and what do I believe about the world? Goal-based: which actions will reach my goal? Utility-based: which action gives the best overall value? Learning: how can I get better from experience?*

Part 7. Applying the Concepts to Scenario Questions

Your scenario-based questions (hospital delivery robot, agricultural drone, warehouse robot and parcel inspection system) all ask for the same set of skills. The scenarios contain deliberate clue sentences. This part explains how to read those clues. It teaches the method, so that you can write the answers yourself in your own words.

7.1 Choosing the agent type from the clues

- If the system follows fixed rules and reacts only to what it senses now, with no memory of earlier cases, it is a **simple reflex agent**.
- If it keeps an internal map or stored readings to deal with things it cannot currently see, but still just applies rules, it is a **model-based reflex agent**.
- If it searches or plans sequences of actions to reach a goal and accepts any route that works, without scoring them, it is a **goal-based agent**.
- If it assigns numerical values to outcomes, combines them with probabilities and picks the highest expected score, it is a **utility-based agent**.
- If it records feedback, improves its estimates or behaviour over time, and has parts such as a critic and an exploration component, it is a **learning agent**.

Also look for statements such as "does not improve from previous missions". They are written to rule out the learning agent. Statements such as "does not rank routes using a numerical score" are written to rule out the utility-based agent.

7.2 Model-based reflex agent versus goal-based agent

Both may keep an internal model. The difference is what they do with it. A model-based reflex agent uses its model only to work out the current situation and then applies a rule that maps the situation straight to an action. A goal-based agent uses its model to imagine *what would happen* if it took different sequences of actions in the future, and then picks a sequence that reaches the goal. The goal-based agent looks ahead; the reflex agent does not.

7.3 Classifying the environment: how to justify

Never write only the label. Each classification needs a one-sentence reason taken from the scenario. A good pattern is: "The environment is [label] because [specific fact from the scenario]." Useful clue words are as follows.

- **Partially observable**: hidden aisles, parts that cannot be seen until nearby, noisy or late sensors. **Fully observable**: the problem states that sensors report everything required completely and without error.
- **Stochastic**: unpredictable people, changing wind, congestion, noise. **Deterministic**: every action works exactly as commanded and the outcome is fully determined.
- **Sequential**: earlier moves affect later options (routes, battery, learning over weeks). **Episodic**: each item is judged independently and earlier decisions do not affect later ones.

- **Dynamic**: things move or change while the agent decides (people, trolleys, other robots, wind). **Static**: the object cannot change while being inspected and the conveyor is stopped.
- **Continuous**: position, speed, distance, weight and dimensions are measured on a smooth scale, so a physical robot is largely continuous. **Discrete**: decisions such as accept, reject or manual inspection, or a menu of actions, are separate choices. A single scenario can contain both: discrete decisions based on continuous measurements.
- **Single-agent or multiagent**: if other robots or humans also make decisions that affect the outcome, it is multiagent.

7.4 Choosing among the four views of AI

Each scenario ends with a sentence about the designers' purpose, such as "the robot is not designed to reproduce human thought processes or human mannerisms". This sentence eliminates the two human-centred views (thinking humanly and acting humanly). Then decide between thinking rationally (correct logical reasoning is the goal) and acting rationally (choosing actions that best achieve goals). When the scenario judges the system by correct and safe behaviour or good decisions under uncertainty, the best match is usually **acting rationally**. When the system simply applies fixed logical rules to reach correct conclusions, you may discuss **thinking rationally**, but read the stated objective carefully. Note also that following fixed rules does not prove that a system thinks like a human; it only shows that it applies rules correctly.

7.5 Formulating a search problem

A goal-based agent solves a task by **search**. To formulate a task as a search problem, define five things.

- **Initial state**: where the agent starts (for example, the robot at the pharmacy with the assigned ward).
- **Actions**: what the agent can do in a state (move forward, turn, stop, request a door to open).
- **Transition model**: what state results from each action.
- **Goal test**: how to check whether the goal is reached (for example, the robot is at the assigned ward and the container is delivered safely).
- **Path cost**: a number that measures how costly a solution is, such as total distance, travel time or energy used.

A path-cost measure improves route selection because the basic search only finds *a* route that works. With a cost measure, the agent can compare all working routes and choose the cheapest one, for example the shortest or the fastest, instead of accepting the first one it finds.

7.6 Utility, expected utility and gathering information

Maximizing expected utility means choosing the action with the best average value, weighing each outcome by its probability. It does not guarantee success, because the unlikely bad outcome can still happen. Collecting another observation can be better than acting immediately when the current estimate is very uncertain and a wrong action would be costly, for example wasting scarce water on a spot that may not actually be dry. A small extra reading costs little and can raise the expected value of the final decision.

7.7 Learning agent: exploration, benefit and risk

In a learning agent, the performance element is the part that chooses and executes actions. The critic is the part that evaluates results using the reward or performance standard. The learning element is the part that updates estimates and behaviour using that feedback. The problem generator is the part that proposes exploratory actions. Exploring a less-used option can benefit the agent by revealing better routes or information it would never otherwise discover. The risk is that the unfamiliar option may be slower or less safe, so exploration must stay within fixed safety limits.

7.8 Observability and internal state

If sensors become noisy, the current reading is no longer a trustworthy picture of the world, so the environment becomes effectively partially observable. The agent then needs internal state, such as stored past readings, to combine several readings and estimate the true value. However, merely storing readings does not make an agent a learning agent. Learning requires that the agent *improves its behaviour or estimates from experience using feedback*. An agent that only remembers and combines readings, but never changes how it decides, is a model-based reflex agent, not a learning one.

Part 8. Quick Revision Sheet

Key definitions in one line each

- **Agent**: perceives through sensors, acts through actuators.
- **Percept**: what the agent senses at one moment. **Percept sequence**: the full history of percepts.
- **Agent function**: the mapping from percept sequences to actions.
- **PEAS**: Performance measure, Environment, Actuators, Sensors.
- **Rational agent**: does the best action given what it knows. It is not the same as omniscient.
- **Observable versus known**: observable is about the current state; known is about the rules of the world.
- **Episodic versus sequential**: independent decisions versus decisions that affect the future.
- **Static, dynamic, semidynamic**: the world does not change, changes, or only the score changes with time.
- **Deterministic, stochastic, strategic**: fully determined outcome, random outcome, or uncertainty caused by other agents.
- **Discrete versus continuous**: countable separate values versus smooth ranges.

Agent types in one line each

- **Simple reflex**: if-then rules on the current percept only.
- **Model-based reflex**: rules plus an internal state to handle what is hidden.
- **Goal-based**: searches or plans action sequences to reach a goal.
- **Utility-based**: chooses the action with the highest expected utility to balance trade-offs.
- **Learning**: improves from experience using a performance element, critic, learning element and problem generator.

Four approaches to AI in one line each

- **Thinking humanly**: model how the human mind works.
- **Acting humanly**: behave so that people cannot tell you from a human (Turing Test).
- **Thinking rationally**: reason correctly using logic.
- **Acting rationally**: choose the actions that best achieve the goals. This is the focus of modern AI.

Study tip

When you meet any new scenario, work in this order: underline the clue sentences, build the PEAS description, decide the agent type from how decisions are made, classify the environment with a reason for each label, and finally connect the designers' stated purpose to one of the four views of AI. If you follow this order consistently, every scenario question becomes a matching exercise between clues and concepts.

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